

Agencies such as CSIR, DST, DBT, ICMT, DRDO, ISRO, DAE and other national labs help post-graduate (PG) students with their final thesis or research projects through their established infrastructures. Certain institutions have started the nanoscience and nanotechnology programmes in collaboration with international universities. Students spend one or two semesters abroad as a part of the mandate of these collaborations.

PhD programmes are also running via such collaborations, in addition to independent guidance courses in universities and institutions.

The government of India has taken the following initiatives to support nanoscience and nanotechnology activities:

1. Initiatives such as DST-Nanomission (nano-biotechnology activities) through DBT, ICMR and CoE in Nanoelectronics by MeitY support nanoscience, nanotechnology, nano-biotechnology and nanoelectronics activities.

2. Eighteen sophisticated analytical instruments facilities (SAIFs) established by DST across India play a major role in advanced characterisation and synthesis of nano-materials for various applications.

3. CoE in Nanoscience and Nanotechnology established by DST-Nanomission helps research and PG students in various thrust areas.

4. Thematic units of excellence (TUEs) for various areas of nanoscience and nanotechnology play a major role in product-based research to sup-

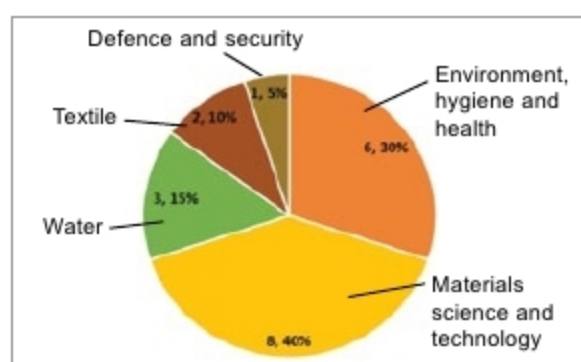


Fig. 1: 20 transfers of technologies

port nanotechnology.

5. Visveswaraya PhD fellowships offered by MeitY supports various nanotechnology activities in the country.

6. INSPIRE scheme supports research fellows to work in interdisciplinary nanotechnology, nanoscience and nano-biotechnology areas.

7. DST-Nanomission supports more than 20 PG teaching programmes to create a baseline for nanoscience and nanotechnology in India, out of about 70 PG programmes currently running in India.

Nanotechnology courses and academic programmes

Out of nearly 70 nanotechnology courses in India, South India occupies around 50 per cent of the institutions. Hence, southern India contributes the highest to capacity-building in nanoscience and nanotechnology.

Seventeen programmes have been funded by the government of India, including 10 from south India.

National Nanotechnology Initiative through Nanomission has impacted the following industries to

tie up with academia and R&D labs to develop products and processes:

1. NFMTC - IITM and MCRC and Orchid Pharma

2. Nanotech Centre - UoH and Dr Reddy's Lab

3. Centre Innovative Smart Textile-IITD, ARCI and Industries

4. Centre for Pharmaceutical Nanotech - NIPER and Pharma Industries

5. Rubber Nanotechnology - MGU and Apollo Tyres

6. Nanophosphor Application Centre, UoA - Nanotech Corp, USA

Many industries have shown interest in the area of nanotechnology in the last decade, including Tata, Mahindra & Mahindra, Piramal, Intel, Orchid Pharma, Apollo, Dr Reddy's, Resil, Cranes, Panacea, Veeco, BEL, Moser Baer, Insta Power, SBP Aqua Tech and Eureka Forbes.

The nanotechnology programme in India has been successful through the following initiatives:

DST-Nanomission. DST sponsored 11 units of nanoscience, seven centres and one computational facility. Public-private partnerships and three private institutions played major roles along with 30 central, autonomous, state and deemed universities. Outcomes of the nanoscience and nanotechnology programme is shown in Fig. 1.

Impacting, Research, Innovation & Technology (IMPRINT). Through IMPRINT-I initiative by the government of India (MHRD in association with other ministries), stress has been given to upgrading nano-education and various modules. Two projects have been initiated on nano-education, and 21 projects on nanotechnology and related hardware.

DRDO. It has also focussed on various nanotechnology activities through its establishments like SSPL, DMSRDE, DLJ, NMRL, DMRL, NPOL, HEMRL and DIHAR. DRDO also started CoEs in following areas:

1. Nanotechnology-based sensors for NBC nuclear, biological, chemical (NBC) detection

2. Nano optoelectronic devices
3. ARCHEM-University of Hyderabad

DBT. This is a nano-biotechnology initiative aimed at the intervention of

TABLE II

INTERVENTION OF NANOTECHNOLOGY IN AGRICULTURE AND ALLIED AREAS

National Labs	Universities	Central institutes
• ARI Pune • CIFE Mumbai • CIRCOT Mumbai • CFTRI Mysore • IMMT Bhubaneswar • NIASM Pune • INST Mohali • CIFA Bhubaneswar • DIHAR-DRDO • ICT Hyderabad • CIAB Mohali • IIFPT Tanjavur • NABI Mohali • NIFTEM Sonapat	• University of Allahabad • University of Madras • Madurai-Kamraj University • Thiruvalluvar University Vellore • University of Delhi • ANGRAU Hyderabad • UAS Dharwad • TNAU Coimbatore • MPUAT Udaipur • TERI-Deakin Nano-bio technology Centre • Sathyabama University Chennai	• IIT Bombay • ISI Kolkata • IARI New Delhi